GG-S-00816c
8 December 1965
SUPERSEDING
Int. Fed. Spec. GG-S-00816b
dated June 26, 1963 and
Fed. Spec. GG-S-816a
dated February 27, 1951

INTERIM FEDERAL SPECIFICATION

SUTURE, NONABSORBABLE, SURGICAL (NONMETALLIC)

This interim Federal specification was developed by Defense Personnel Support Center, Directorate of Medical Materiel, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, based on currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim Federal Specification as a valid exception to Federal Specification GG-S-816a and Interim Federal Specification GG-S-00816b.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sterile, nonabsorbable, surgical sutures, with and without needles attached.

1.2 Classification.

1.2.1 Sutures. Sutures shall be of the following compositions, types, colors, sizes, lengths and classes, as specified (see 6.1):

Composition A - Silk
  Type I - Regular (braided and uncoated)
  Type II - Dermal (twisted and protein-coated)
Composition B - Cotton (twisted)
Composition C - Nylon (monofilament)
Composition D - Polyester fiber (braided)

Color (a) - White
Color (b) - Black
Color (c) - Dark and distinctive
Color (d) - As specified

FSC 6515

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Size - 10-0
Size - 9-0
Size - 8-0
Size - 7-0
Size - 6-0
Size - 5-0
Size - 4-0
Size - 3-0
Size - 2-0
Size - 1-0
Size - 1
Size - 2
Size - 3
Size - 4
Size - 5
Size - 6
Size - 7

Length - 18 inches
Length - 24 inches
Length - 30 inches
Length - As specified

Class 1 - Unarmed
Class 2 - Single armed
Class 3 - Double armed

1.2.2 Needles. Needles shall be of the following classes and points, as specified (see 6.1):

Class A -3/8 circle
Class B - 1/2 circle
Class C - Half-curved
Class D - Straight

Point 1 - Cutting edge (apex inside)
Point 2 - Reverse cutting edge (apex outside)
Point 3 - Taper
Point 4 - As specified

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2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATIONS

CCC-T-191 Textile Test Methods
PPP-B-566 Boxes, Folding, Paperboard
PPP-B-585 Boxes, Wood, Wirebound
PPP-B-601 Boxes, Wood, Cleated-Plywood
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 Box, Fiberboard
PPP-B-676 Box, Setup

FEDERAL STANDARDS

Fed. Test Method Standard No. 151 Metals, Test Methods

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices, in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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MILITARY SPECIFICATION

MIL-L-10547 Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

MILITARY STANDARDS

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. PHARMACOPOEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York 17, N.Y.)

3. REQUIREMENTS

3.1 Materials. All materials shall be suitable for nonabsorbable sutures and as described in the current edition of the Pharmacopeia of the United States (U.S.P.). Fiber identification shall be conducted in accordance with CCC-T-191.

3.1.1 Silk. Silk shall consist of the thread prepared from the degummed cocoon filaments of glutinous gum which are secreted or spun by the mulberry silkworm, Bombyx mori Linne (Fam. Bombycidae).

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3.1.1.1 Coating. (Type II - Dermal sutures only). Coating for sutures shall be composed of a suitable flexible nonabsorbent protein gel. The finished product shall be of a high grade suitable for sutures.

Note: This coating referes to a surface treatment (encasement) applied to make sutures very smooth and resistant to ingrowth by tissue cells, and not to the treatment (such as a wax impregnation) applied to sutures to render them noncapillary.

3.1.2 Cotton. Cotton suture shall consist of a thread prepared from Egyptian long staple karnak cotton.

3.1.3 Synthetic fibers.

3.1.3.1 Nylon. The nylon shall be made from high grade oriented polyamide, suitable for sutures, having a melting point of not less than 200° centigrade (C.).

3.1.3.2 Polyester fiber (braided). Sutures shall be made from high-grade polyester fiber, having a melting point of not less than 200° centigrade (C.).

3.1.4 Needles. The needles shall be fabricated from AISI type 420 or 440A stainless steel, or a suitable carbon steel.

3.1.4.1 Carbon steel needles may be nickel plated. The plating shall be adherent, and shall not impair the sharpness of the needles.

3.1.5 Packet fluid. The packet fluid shall be a clear, colorless, nontoxic solution. The fluid shall be such as not to have any deleterious effect on the sutures or the needles during prolonged storage.

3.1.6 Plastic (container). Plastic shall be of good quality, and of proper permeability to be suitable for the purpose intended. The plastic shall be nontoxic, and compatible with the contents of the container.

3.1.7 Aluminum foil (container) The aluminum foil shall be of good quality suitable for the purpose intended.

3.1.8 Reels. Reels shall be fabricated from suitable nontoxic material that is compatible with the contents of the container. Any reels used with packet fluid shall not dissolve or disintegrate in the fluid, or cause any discoloration or turbidity thereof.

3.2 Performance and product characteristics. The sutures and needles shall be suitable for use in the practice of surgery, and for any specific surgical purpose specified (see 6.1).

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3.3 Description. Except as herein specified, the sutures shall conform to all of the requirements for nonabsorbable sutures in the current edition of the U.S.P. Each assembly shall consist of the number of strands of the composition, type, color, size, length, and class of suture specified (see 6.1). The suture shall be sterile, evenly coiled or wound on a reel in such a manner as to permit its ready dispensing, and shall be sealed in the container specified (see 3.4). Needles on armed sutures shall be arranged so that they cannot accidentally puncture the container during normal handling.

3.3.1 Suture composition.

3.3.1.1 Composition A (silk). Composition A sutures shall be of silk, as specified in 3.1.1. All silk sutures shall be treated to be noncapillary, and when tested as specified in 4.4.2, there shall be no evidence of capillary action. Type I silk sutures shall be comprised of braided silk in accordance with requirements for braided silk (U.S.P. "Class I") contained in the current edition of the U.S.P. Type II silk sutures shall be comprised of coated twisted silk in accordance with requirements for coated and twisted silk (U.S.P. "Class II") contained in the current edition of the U.S.P. The coating shall be as specified in 3.1.1.1, and shall result in a smooth, nonporous finish. Packet fluid shall be used with type II sutures.

3.3.1.2 Composition B (cotton). Composition B sutures shall be of twisted cotton, as specified in 3.1.2. Cotton sutures shall not be treated to reduce capillarity, and shall be in accordance with requirements for cotton sutures ("Class II") contained in the current edition of the U.S.P.

3.3.1.3 Composition C (nylon). Composition C sutures shall be of monofilament nylon, as specified in 3.1.3.1. The nylon sutures shall be in accordance with requirements for sutures of synthetic fibers ("Class I") contained in the current edition of the U.S.P. The sutures shall have a smooth, nonporous finish.

3.3.1.4 Composition D (polyester fiber). Composition D sutures shall be of braided polyester fiber, as specified in 3.1.3.2. The sutures may be coated (or impregnated) with fluorocarbon resin. The sutures shall be noncapillary, and shall be in accordance with the requirements for sutures of synthetic fibers ("Class I") contained in the current edition of the U.S.P.

3.3.2 Suture classes. Class 1 (unarmed) suture shall be supplied without needles, class 2 (single armed) suture shall be supplied with one eyeless needle securely attached to one end of the suture, and class 3 (double armed) suture shall be supplied with two eyeless needles, one securely attached to each end of the suture. The needles shall be as specified in 3.3.5.

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3.3.3 Size. Sutures shall be of the size (diameter) specified in accordance with size designations and requirements, contained in the current edition of the U.S.P., for nonabsorbable sutures.

3.3.4 Length. Each strand of suture shall be of the length specified with a tolerance of plus 10 percent or minus 5 percent. Each suture strand shall be of one continuous piece.

3.3.5 Needles. Class, point, and dimensions of the needles shall be as specified (see 6.1). They shall be fabricated of material as specified in 3.1.4. The needles shall be of the eyeless type, having smooth surfaces to eliminate tissue drag, and to minimize tissue trauma. The needles may have longitudinal ribs, on the shank portion only, to facilitate gripping by a suture needle holding forceps, provided the ribs are not excessively sharp, or deep cut so as to seriously weaken the needles. All needles shall be sharp, not blunt, bent, or hooked, and shall be free of feathered edges, burs, cracks, pits, or other defects, when examined under 10 power (10X) magnification. When cutting edge needles (point 1) are specified, the apex of the triangular cross-section of the point shall face the inside of the needle curvature. When reverse cutting edge needles (point 2) are specified, the apex shall face the outside of the needle curvature. The needles shall have a smooth, bright finish.

3.3.5.1 Hardness. When tested as specified in 4.4.3, the needles shall have a Rockwell hardness of C45 minimum, or its equivalent as shown in the hardness conversion tables of Federal Test Method Standard 151. The hardness test shall be performed in the vicinity of the needle point, at least 1/4 inch beyond the proximal end (i.e., from where the suture emerges).

3.3.5.2 Ductility. When the needle is submitted to the bend test specified in 4.4.4, no breakage shall occur.

3.3.5.3 Corrosion resistance. (Stainless steel needles)

3.3.5.3.1 Resistance to boil test. When tested as specified in 4.4.5.1, the needles, upon examination, shall not show more than 2 percent corrosion of each needle surface.

3.3.5.3.2 Resistance to copper-sulfate test. When tested as specified in 4.4.5.2, the needle shall show no plating of copper as a result of the test. Dulling of the polished surface, which may be caused by the copper-sulfate solution, shall not be considered as noncompliance with this specification.

3.3.6 Reels. Reels, if used, shall be designed so as to permit the suture to be neatly wound thereon, and to permit its ready dispensing.

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3.4 Immediate container. The specified sterile suture (including packet fluid, needles, and reel, where applicable) shall be placed in a double container. The completed, sealed, individual package shall consist of the sterile suture in the inner container enclosed, in turn, in an outer container that shall maintain the inner container sterile, through changing temperatures from minus 25° C. to plus 60° C. The package shall protect the needles from corrosion, and shall be so constructed as to be readily opened. The double container shall consist of an inner and outer container fabricated of plastic, aluminum foil, paper laminated to aluminum foil, or suitable combination thereof (see 3.1.6 and 3.1.7). The suture shall be placed in the inner container, with or without packet fluid. Closure of the inner container shall be accomplished by heat sealing. The container shall be fabricated in such a manner, and of such material, as to maintain the sterility of the suture. This container, in turn, shall be placed in an outer container fabricated of materials as specified above. Closure of the outer container shall be accomplished by heat sealing. The outer container shall be so designed as to permit ready dispensing of the contents by sterile techniques and without damaging the inner container. The sealed outer container design shall include two free tabs along the top edge. The tabs shall be capable of being readily peeled back along the sealed edges to expose the inner container as shown in figure 1. This shall be done without contaminating the inner container when opening the outer container and removing the inner container. The outer container shall be fabricated in such a manner, and of such material, as to maintain the sterility of the inner container. The inner and outer container shall be tested for sterility (see 3.7) as specified in 4.4.8. The inner container, when tested as specified in 4.4.6.1, shall not burst, or leak (indicated by a steady stream of gas bubbles). The inner container shall be capable of being resterilized in a solution of 98.5 percent alcohol and 1 percent formaldehyde for 30 days without its deterioration and without any harmful effects on the suture contained therein. When tested as specified in 4.4.6.2, the outer container shall not leak (including by capillary action) so as to effect the sterility of its contents. The outer container need not comply with this leakage test if 12 sutures, in their immediate containers, are heat-sealed in a 4 mil polyethylene bag prior to their insertion into the required unit package.

3.5 Security of needle attachment, (needle pull-out). Sutures, armed with eyeless needles, shall be submitted to the security of needle attachment test specified in 4.4.7 (both ends for class 3).

3.6 Control (lot) numbers. The manufacturer shall maintain control numbers in order to identify each lot. The control number shall be entered in the manufacturer's original and permanent records of sterilization and shall also appear on each shipment as hereinafter specified. The manufacturer's records shall be open for Government inspection. When sterilization is accomplished in two steps, only one control number need be indicated. However, the manufacturer shall maintain records which will identify the contents with both sterilization lots.

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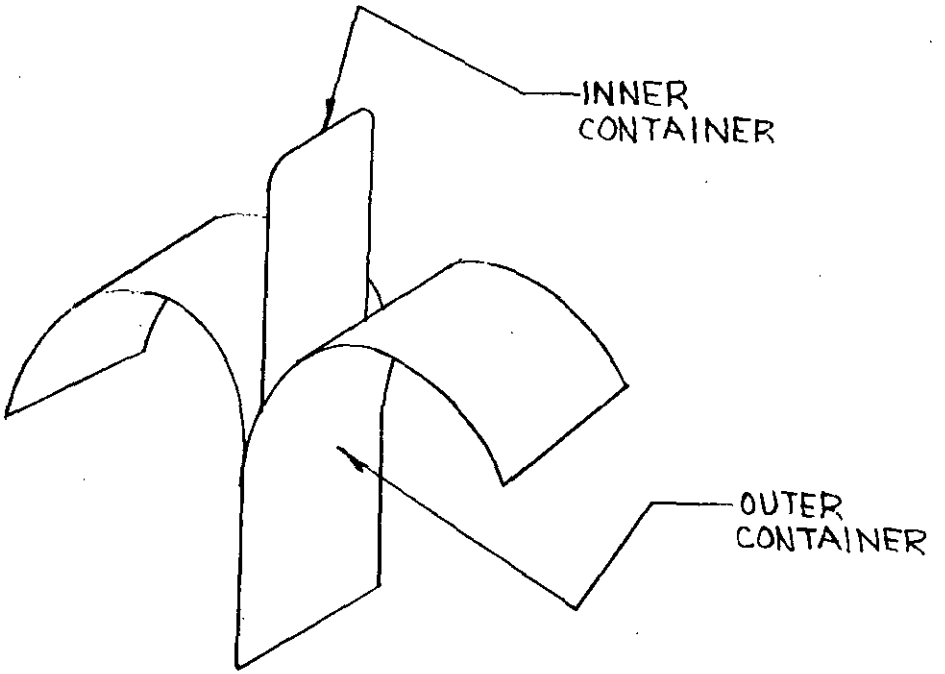


FIGURE 1

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3.7 Sterility. The contents of the outer container (including the inner container), and the contents of the inner container (including suture, needles, reel and fluid) shall be sterile. The suture and the inner container shall comply with requirements for sterility as contained in the current edition of the U.S.P., when tested as specified in 4.4.8.

3.7.1 Certificate of sterility. When inspection is made at destination, a certificate of sterility, listing all lot or control numbers, stating that samples from each sterility lot have been tested in accordance with U.S.P. method and have been found to be sterile, shall be submitted by the contractor to the Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, Attention: Quality Control Branch. A copy of the certificate shall be forwarded with each shipment to the consignee.

3.8 Instructions. Each unit package shall be provided with instructions describing the proper techniques for dispensing the sterile sutures from the containers. In addition, information cautioning the user against the use of chemicals, anticorrosive agents, solutions, etc., that would be harmful to the inner container shall be included, along with a statement (when applicable) indicating the maximum number of days that inner containers may remain in sterilization solutions without harm to the sutures.

3.9 Labels. Each inner container shall be provided with a commercial label identifying the suture as to its composition, color, capillarity, size, length, needle class (or illustration of class), needle point, the number of needles or the legend "single armed" or "double armed", name of contractor, the legends "sterile", "nonboilable", and "braided", and other requirements of statute. As an alternate, the registered trademark may be substituted for name of contractor. Reels may be used as labels. The label shall be visible through the outer container.

3.10 Workmanship. Workmanship shall be first class throughout. Sutures shall be smooth, free from kinks, splinters, fraying, unraveling, cuts or knots. Needles shall be as described in 3.3.5. The entire assembly, including sutures, needles, reels, fluid and container, shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. Except where otherwise indicated, the inspection lot shall be determined in accordance with MIL-STD-105, and shall be based on the unit of product being one strand of suture (end item). For double armed sutures, the inspection lot for the needles only shall be based on one individual needle.

4.2.2 Sampling for examination. A random sample of sutures shall be selected from each lot in accordance with MIL-STD-105. The inspection level and Acceptable Quality Level (AQL) shall be as indicated in table I.

Table I Sampling for examination

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S2 *	1.0
Minor	II	2.5

* If acceptance number is not exceeded, lot shall be acceptable. If rejection number is exceeded, inspection at level II may be performed to determine final rejection or acceptance.

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4.2.3 For tests (including diameter). Sampling for tests, and diameter, shall be conducted as specified in table II. Tests for the suture proper shall be performed on the finished sterilized suture.

Table II. Sampling for tests 1/

Component	Characteristic	Rqmt.	Test procedure	Inspection level 1/	Minimum number of samples to be tested	Acceptance number
Needle	Chemical analysis	3.1.4	4.4.1	2/	2/	0
Suture	Capillarity	3.3.1.1	4.4.2	--	2	0
Needle	Hardness	3.3.5.1	4.4.3	S-1	3	0
Needle	Ductility	3.3.5.2	4.4.4	S-1	3	0
Needle	Boil test	3.3.5.3.1	4.4.5.1	S-2	3	0
Needle	Copper sulfate test	3.3.5.3.2	4.4.5.2	S-2	3	0
Inner and outer containers	Leakage test	3.4	4.4.6	S-2(3/)	4	0
Suture & needle	Security of needle	3.5	4.4.7	4/	4/	0
As applicable	All other tests (including diameter)	As applicable		5/	5/	0

1/ Except as otherwise indicated, sampling shall be conducted as specified in MIL-STD-105.

2/ A sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.

3/ The inspection lot for container leakage test shall not exceed 20,000 containers, or one day's uninterrupted production.

4/ Ten units shall be selected at random from each lot for testing.

5/ Sampling for each test shall be conducted in accordance with the current edition of the U.S.P.

4.2.4 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

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4.3 Examination, The sutures and needles shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Inspection, however, shall not be restricted to defects listed in table III. Any sutures (or needles) in the sample containing one or more defects shall be rejected, and if the number of defective sutures (or needles) in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical	None defined
Major A	
101	Holes or tears in inner or outer container.
102	Opening in seal of inner or outer container.
103	Required label for inner container not visible, incorrect, not legible, incomplete, or missing.
Major B	
151	Immediate container does not readily, or properly, open.
152	Suture not flexible.
153	Surface of suture not smooth and free of fraying, unraveling, kinks, cuts, or knots.
154	Suture strand not one continuous piece and of length specified.
155	Suture, and contents of inner and outer container not free of foreign matter or discoloration.
156	Suture entangled, snarled, knotted, or other defects which prevent its ready dispensing from reel or coil.
*157	Needle missing or detached from suture.
*158	Needle not firmly attached to suture.
*159	Needle blunt or bent, not sharp, and not free of feathered edges, burrs, or cracks (examined under 10X magnification).
*160	Needle generally pitted or otherwise not smoothly or symmetrically finished.
*161	Dimensions of needle not within tolerances specified.
Minor	
201	Inner container (aluminum foil only), not marked to indicate proper point for opening.
202	Instructions incorrect, not legible, incomplete, or missing.

*Applicable to armed suture only.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Except as hereinafter specified, all tests shall be conducted as specified in the current edition of the **U.S.P.** Tensile strength test shall be performed on the Incline Plane Tester described in the current **U.S.P.**

4.4.1 Chemical analysis. Chemical analysis of needles shall be determined as specified in method 111 or 112 of Federal Test Method Standard No. 151.

4.4.2 Capillarity test. At least two 3- or 4-inch lengths of the suture shall be boiled in distilled water in a glass container for three successive periods of 20 minutes each. After the third boiling, they shall be removed, laid flat, and conditioned for 8 hours ± 10 minutes in an atmosphere having a relative humidity of 65 ± 2 percent at a temperature of 21° ± 1.1° C. (70° ± 2° F.). After conditioning, a piece of white silk thread shall be knotted on one end of the suture and the ends of the knot cut close. The suture shall then be suspended in a dye solution of 0.5 percent methylene blue in such a manner that the white silk knot on the strand under test projects 3/8 inch above the top of the liquid. After a period of 24 hours any evidence of the dye being carried up the strand and through the white silk knot, where it is readily visible, will be considered capillary action.

4.4.3 Needle hardness test. Hardness shall be determined in accordance with Method 243 (Rockwell) or Method 244 (Diamond Pyramid) of Test Method Standard No. 151. As an alternate, scratch test method for hardness determination shall be acceptable.

4.4.4 Needle ductility test. The needles shall be tested as described in Method 231 of Test Method Standard No. 151 for longitudinal bend test. The needles shall be bent through an angle of 45° over a die, or round edge, of a diameter equal to the wire size of the needle tested.

4.4.5 Needle corrosion tests.

4.4.5.1 Boil test. Sample needles shall be scrubbed with soap and warm water, rinsed in distilled water, and dried. They shall then be boiled for 30 minutes in distilled water in a glass beaker. They shall remain submerged for 48 hours, then allowed to dry and examined under 4 to 7 power (7X) magnification.

4.4.5.2 Copper-sulfate test. Sample needles shall be scrubbed with soap and warm water, rinsed in hot water, dipped in ethyl alcohol (95 percent), and allowed to dry. They shall then be completely immersed in the copper sulfate solution at a temperature of 65° ± 2° Fahrenheit (F.) for 6 minutes and shall then be washed with a cloth saturated with fresh water. The copper sulfate solution shall be made up as follows:

Copper sulphate
($CuSO_4 \cdot 5 H_2O$) - - - 4.0 grams

Sulphuric acid (H_2SO_4)
(Specific gravity 1.84) - 10.0 grams

Water (H_2O) - - - - - - - - 90.0 milliliters

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4.4.6 Immediate container leakage tests.

4.4.6.1 Inner container leakage test. The inner containers shall be conditioned at ambient conditions for at least 4 hours prior to performing the test. A small quantity of water-soluble detergent (approximately 4 grams per gallon of water or equivalent amount of wetting agent) may be added to the test water to remove surface-entrapped air. The samples to be tested shall be submerged under water contained in a vacuum vessel from which they can be observed from the outside. The samples shall then be submitted to a vacuum of 8 inches of mercury for 30 seconds. The above test shall be repeated after inverting the samples. While under vacuum, the containers shall be observed for bursting, or leakage indicated by a steady stream or recurring succession of bubbles from any surface or seam. Bubbles which appear on the surface of the container are not to be construed as indication of failure.

4.4.6.2 Outer container leakage test. The outer container shall be completely submerged in water containing a suitable dye for 3 hours.

4.4.7 Security of needle attachment test. Shall be in accordance with the procedure for security of needle attachment test in current edition of U.S.P.

4.4.8 Sterility test. Test for sterility of the inner container shall be performed as follows: The outer container shall not be placed in any bactericidal solution. The outer container shall be opened, and the inner container removed, in accordance with the manufacturer's instructions (see 3.8). The inner container shall then be submitted to sterility test in accordance with the current edition of the U.S.P.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Twelve (12) sutures, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	144 units

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5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-636, type I, class 1, or PPP-B-676. Commercial colors shall be acceptable when carton conforming to PPP-B-676 is utilized. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall bear the following information which is not required to appear in the sequence indicated:

(a) Federal stock number
(b) Item identification
(c) Unit and quantity
(d) Contract or purchase order number
(e) Lot (control) number
(f) Type, capillarity, color and length of suture
(g) As applicable, class (e.g. 3/8 circle) and point (e.g. taper) of needle, when furnished
(h) Composition of packet fluid (when applicable)
(i) Name of contractor; when contractor is not the manufacturer, the name of the manufacturer shall also appear
(j) Information as may be required by Federal and state laws
(k) The legends "STERILE" and "NON-BOILABLE"
(l) Instructions as specified in 3.9

Commercial markings, when utilized, shall be acceptable provided that there shall be no interference with the information herein specified.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number, lot (control) number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) numbers shall be shown.

Note. The quantity and unit shall be designated on interior packages and exterior container as follows:

Unit package: 1 PKG
(of 12 sutures)

Intermediate package: 12 PKG

Exterior container: 144 PKG

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Composition, type, color, size, length, and class of sutures required (see 1.2.1).

(c) Class, point, and dimensions of needle required (see 1.2.2 and 3.3.5).

(d) Specific surgical application for which required (see 3.2).

(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy -MS
Air Force- 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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